PRACTICAL NO 4: Configure IP ACLs to Mitigate Attacks.

Access Control Lists (ACLs)

Network administrators must figure out how to deny unwanted access to the network while allowing internal users appropriate access to necessary services.
Although security tools, such as passwords, callback equipment, and physical security devices are helpful, they often lack the flexibility of basic traffic filtering and the specific controls most administrators prefer.
For example, a network administrator may want to allow users access to the Internet, but not permit external users telnet access into the LAN.
Routers provide basic traffic filtering capabilities, such as blocking Internet traffic, with access control lists (ACLs).
An ACL is a sequential list of permit or deny statements that apply to addresses or upper-layer protocols.
The router examines each packet to determine whether to forward or drop it, based on the conditions specified in the ACL.
Some ACL decision points are:
1) IP source address
2) IP destination addresses
3) UDP or TCP protocols
4) Upper-layer (TCP/UDP) port numbers

ACLs must be defined on a:
1) Per-protocol (IP, IPX, AppleTalk)
2) Per direction (in or out)
3) Per port (interface) basis.
4) ACLs control traffic in one direction at a time on an interface.
5) A separate ACL would need to be created for each direction, one for inbound and one for outbound traffic.
6) Finally every interface can have multiple protocols and directions defined.

An ACL is a group of statements that define whether packets are accepted or rejected coming into an interface or leaving an interface.
1) ACL statements operate in sequential, logical order (top down).
2) If a condition match is true, the packet is permitted or denied and the rest of the ACL statements are not checked.
3) If all the ACL statements are unmatched, an implicit "deny any" statement is placed at the end of the list by default. (not visible)

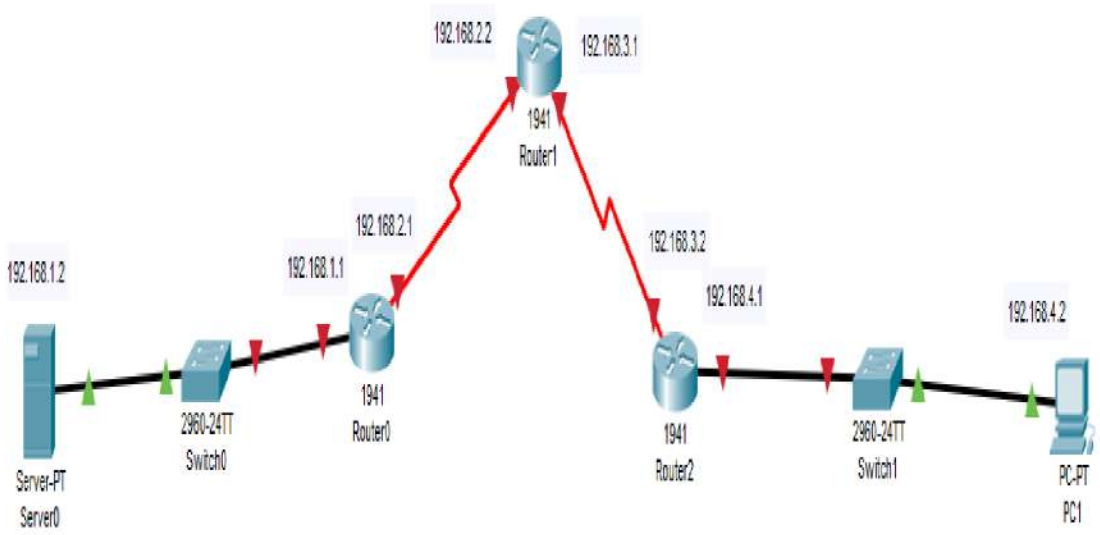
When first learning how to create ACLs, it is a good idea to add the implicit deny at the end of ACLs to reinforce the dynamic presence of the command line.

Standard IP ACLs
- Can only filter on source IP addresses

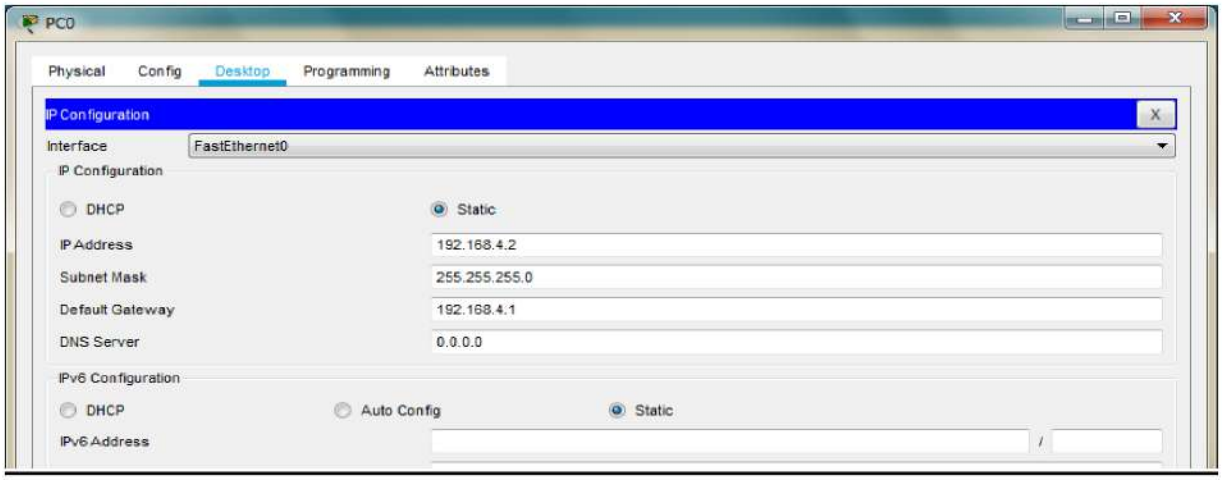
Extended IP ACLs Can filter on:
1) Source IP address
2) Destination IP address
3) Protocol (TCP, UDP)
4) Port Numbers (Telnet – 23, http – 80, etc.) and other parameters

An access list is a sequential series of commands or filters. These lists tell the router what types of packets to: accept or deny Acceptance and denial can be based on specified conditions. ACLs applied on the router's interfaces

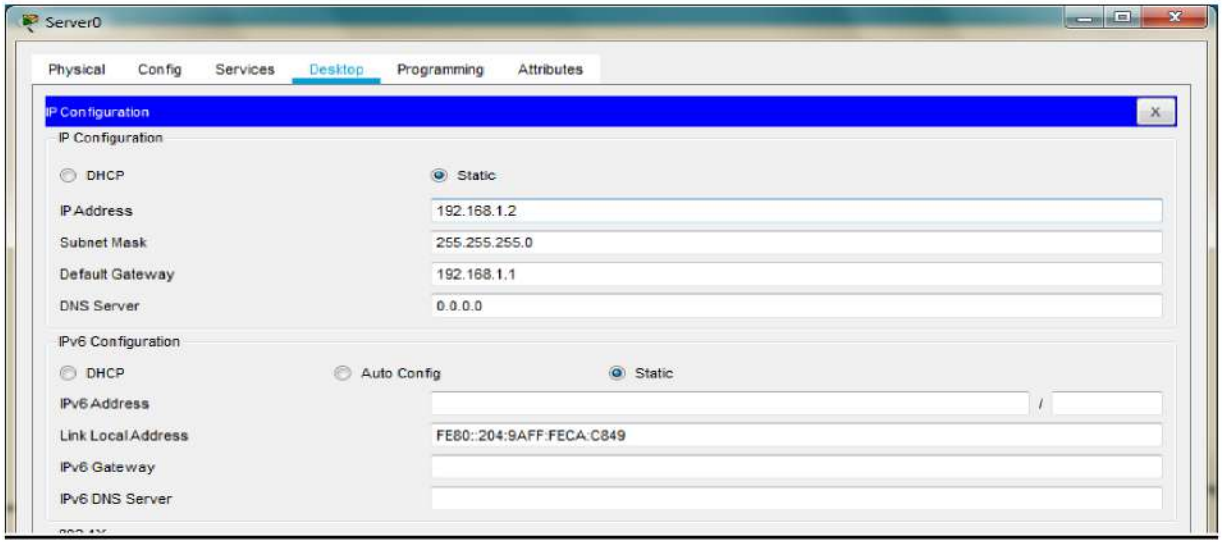
We use the following topology to study the present case



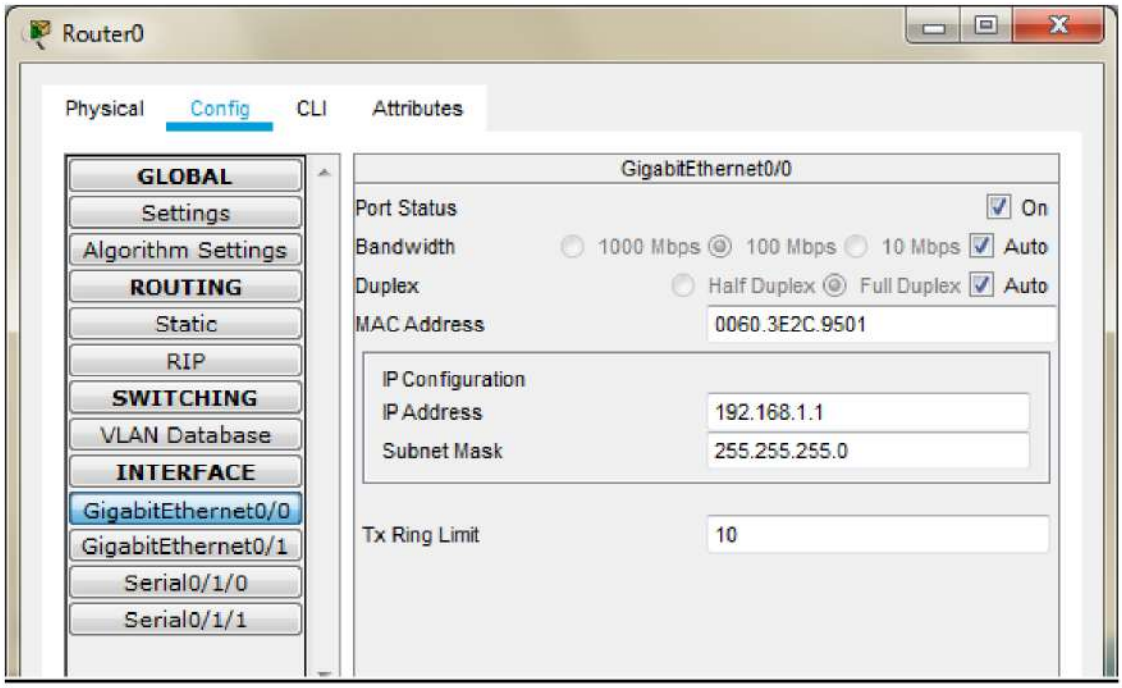
<u>Configuring PC1</u>

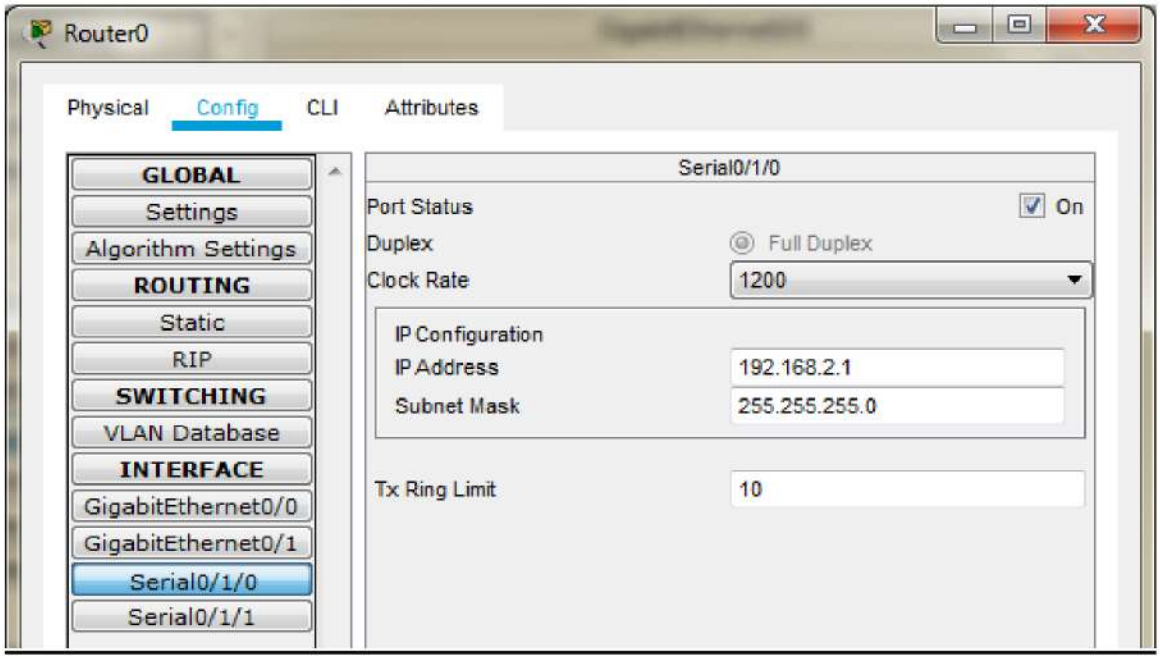


Configuring Server0

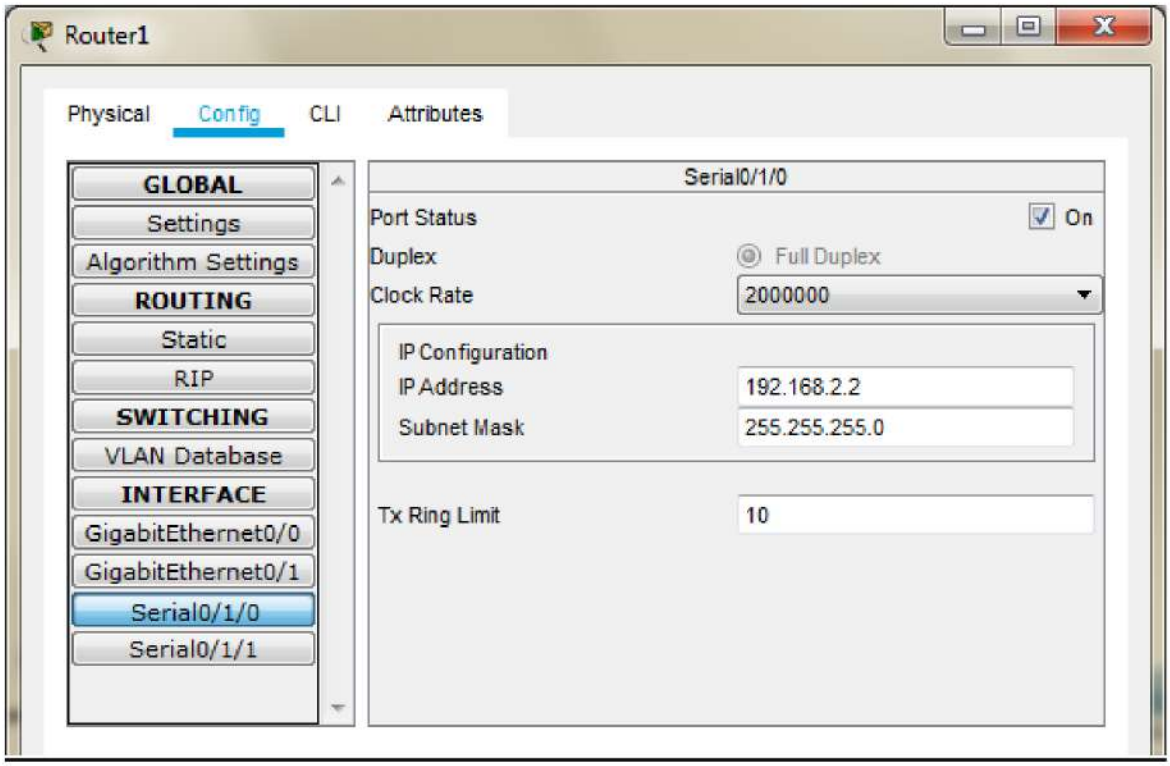


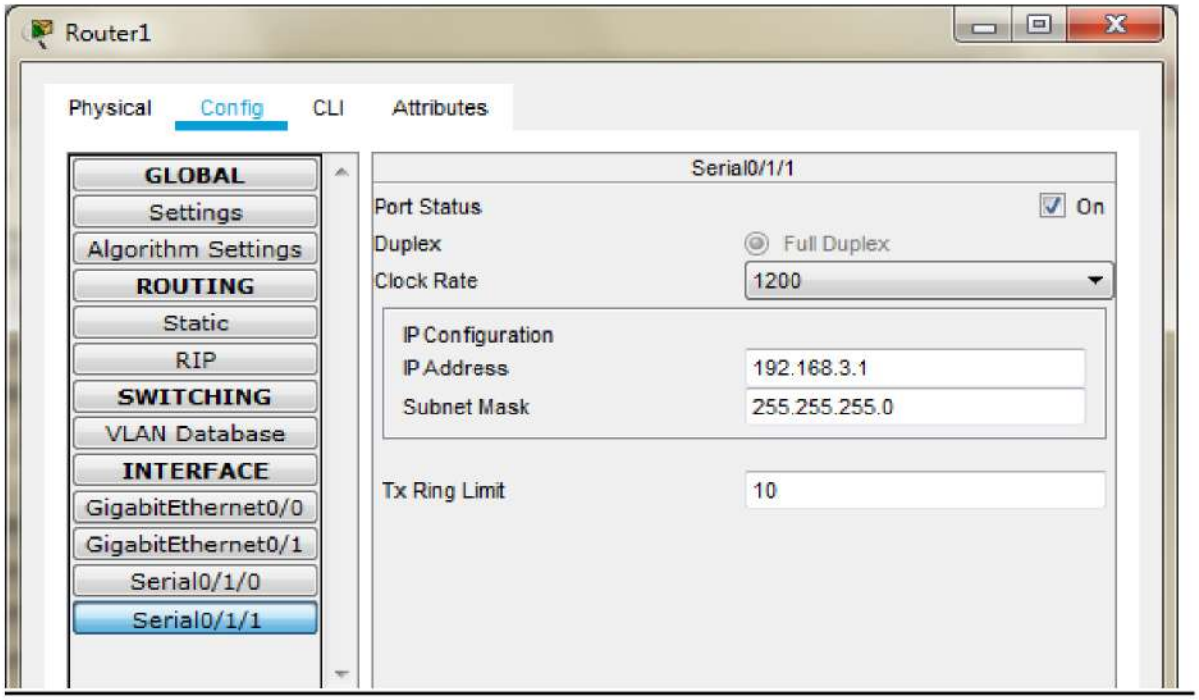
Configuring Router0



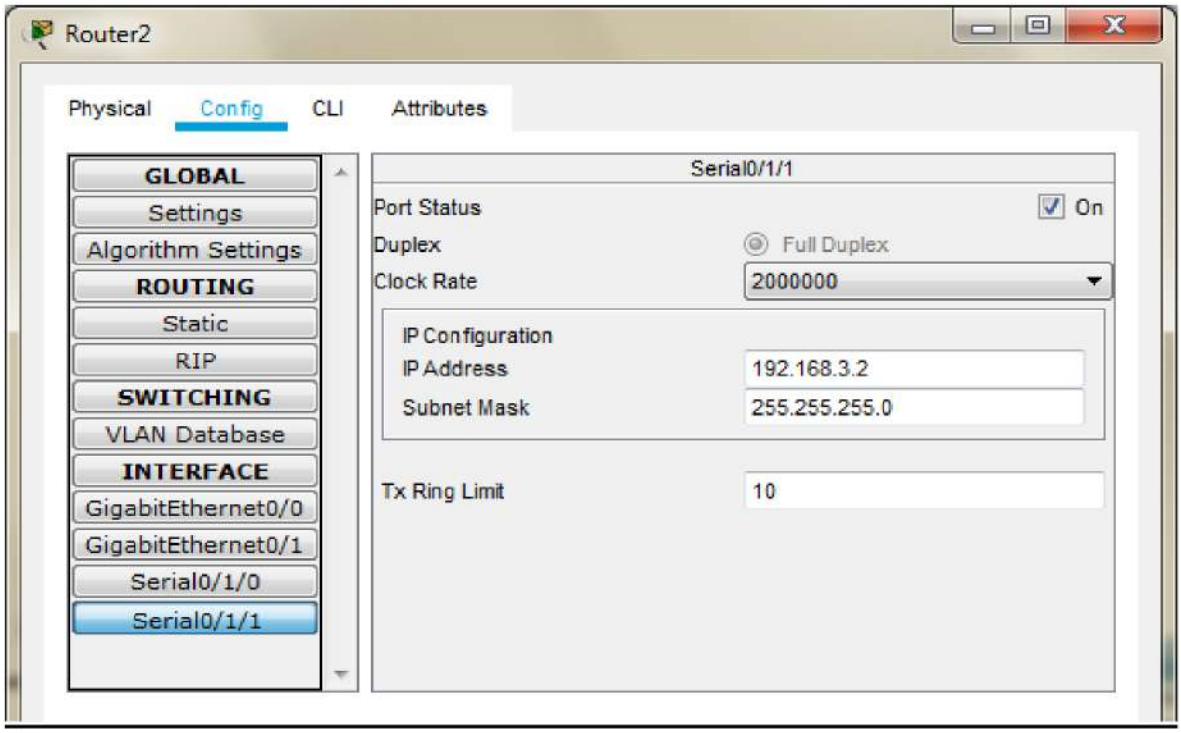


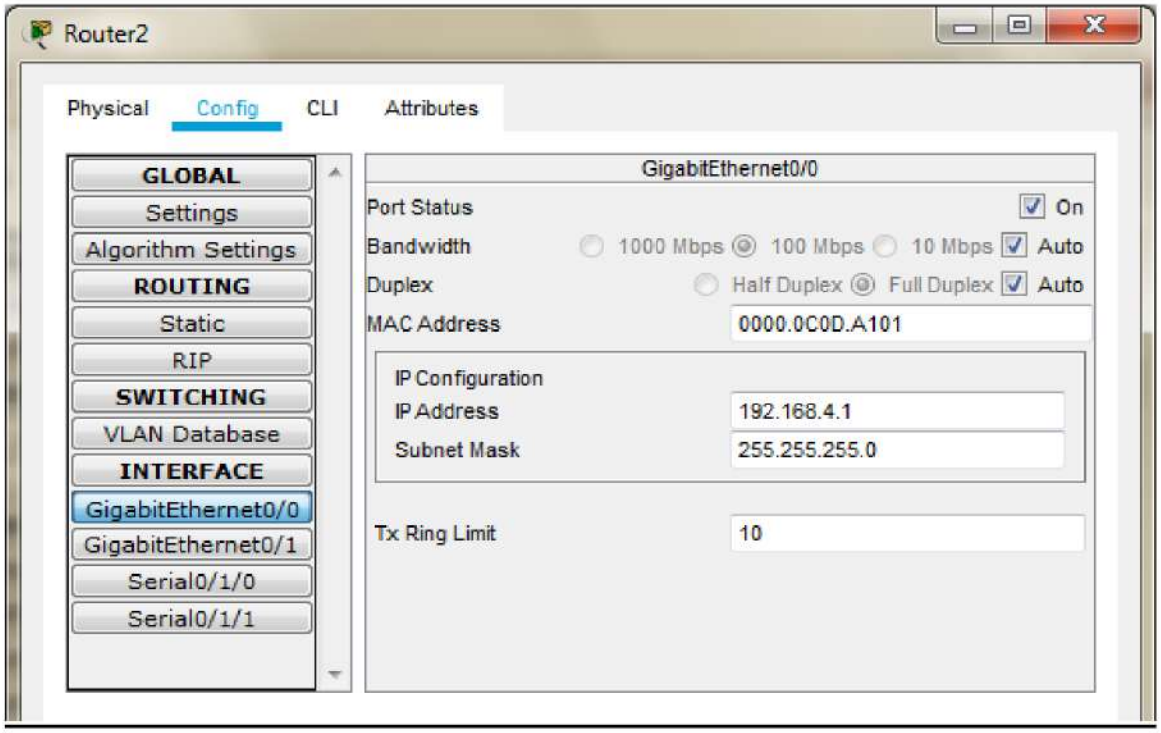
Configuring Router1



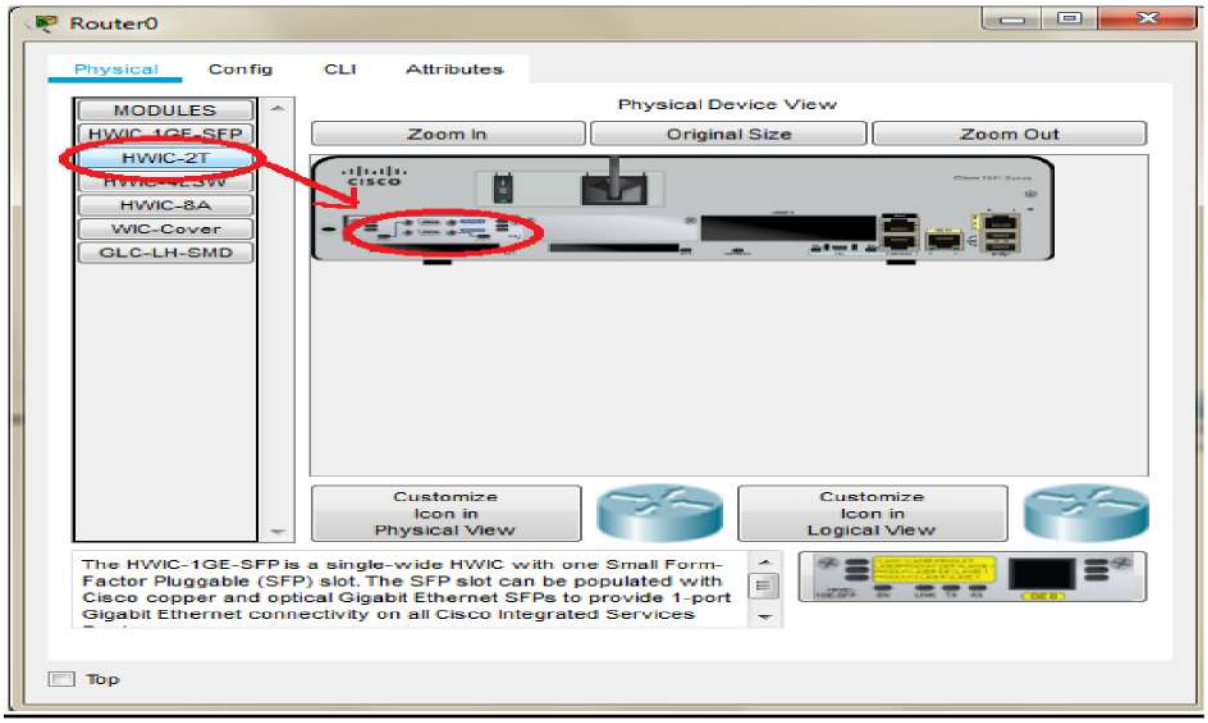


Configuring Router2

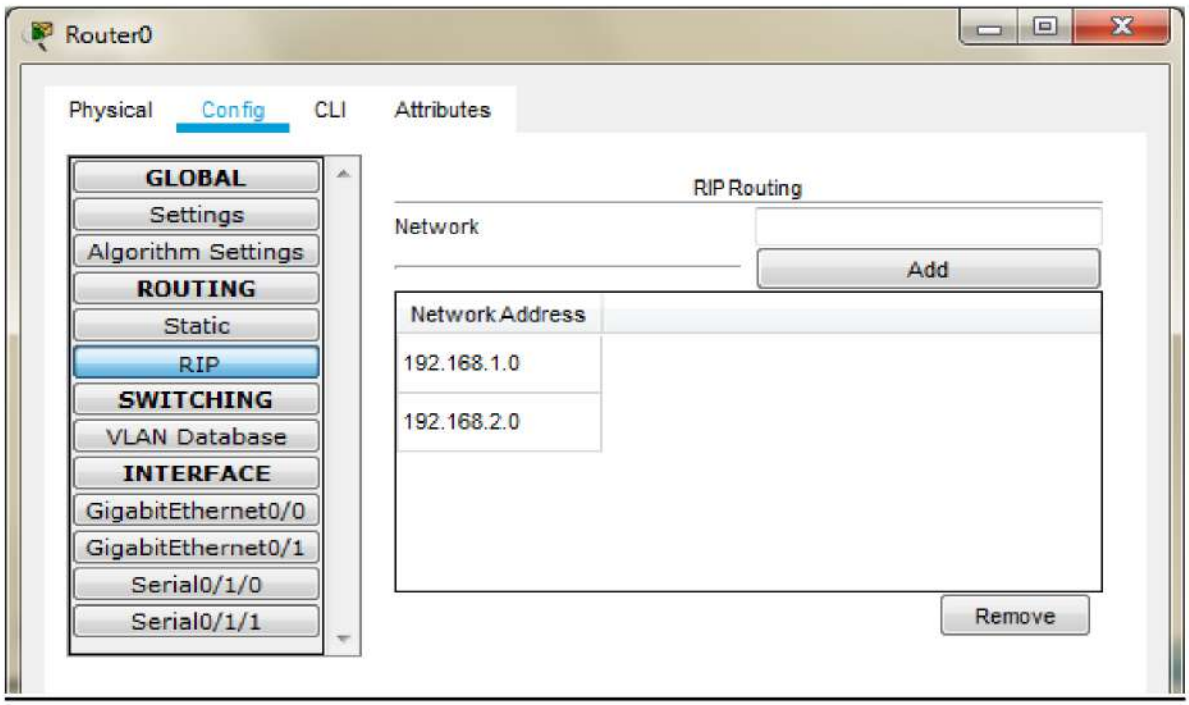


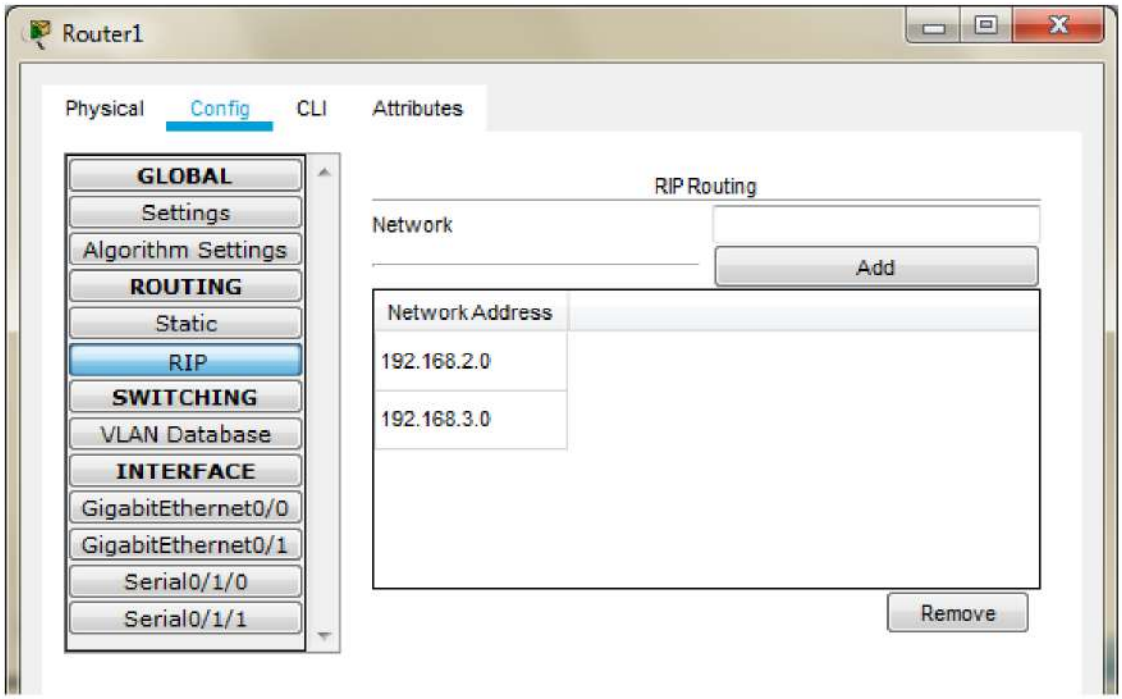


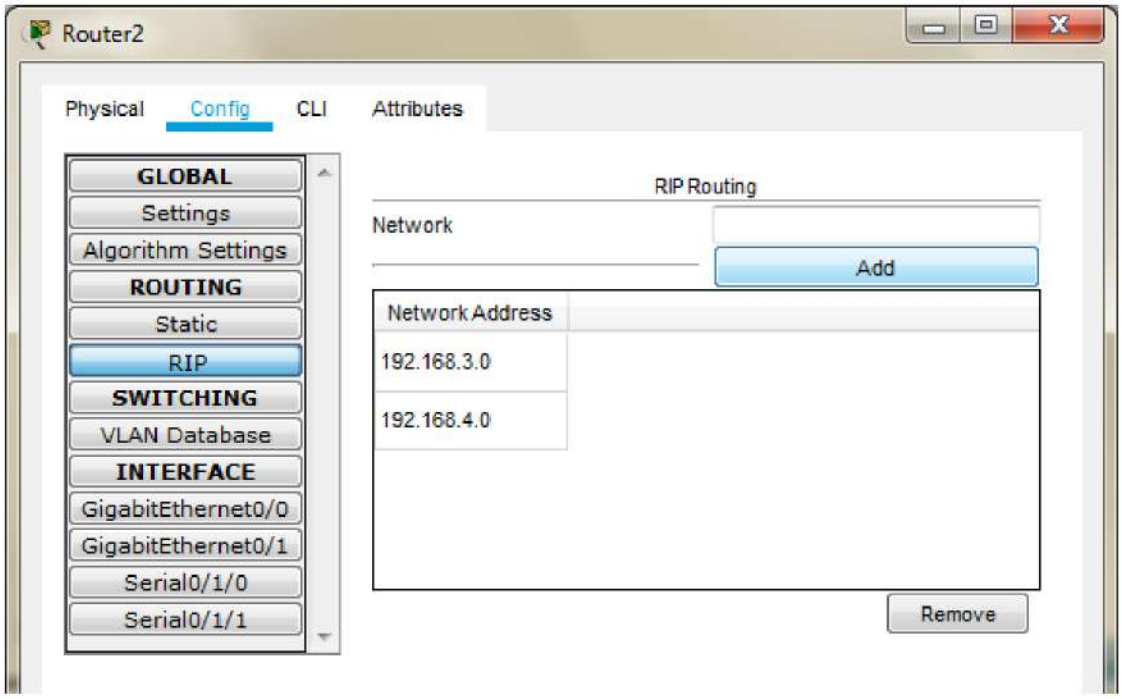
<u>The serial interface in each Router are added as follows</u>



Set the RIP on each Router

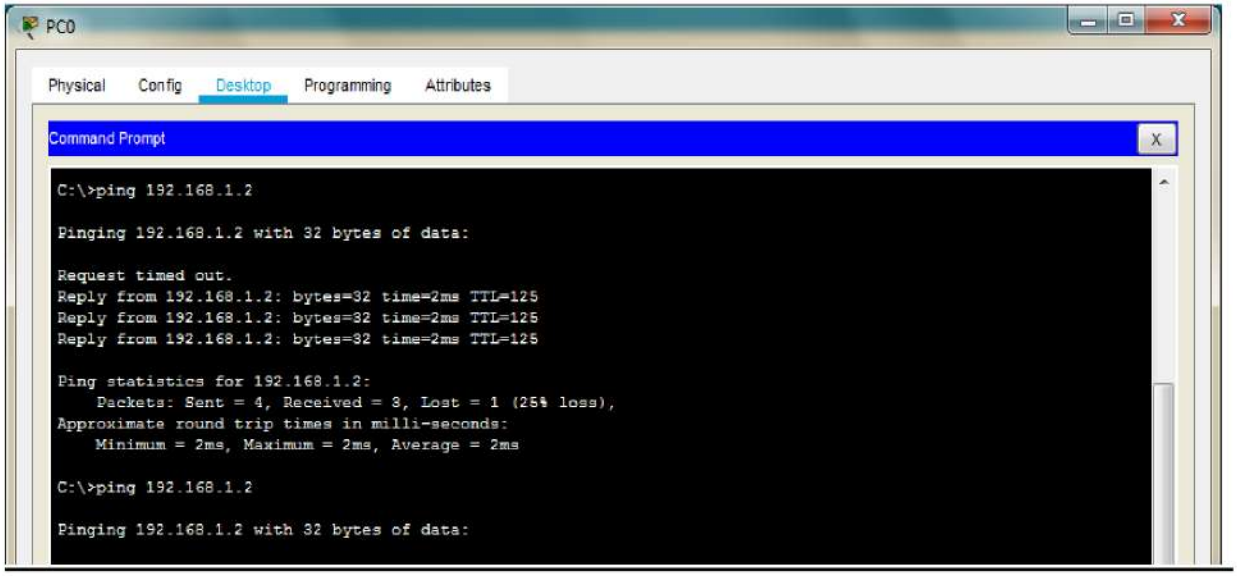




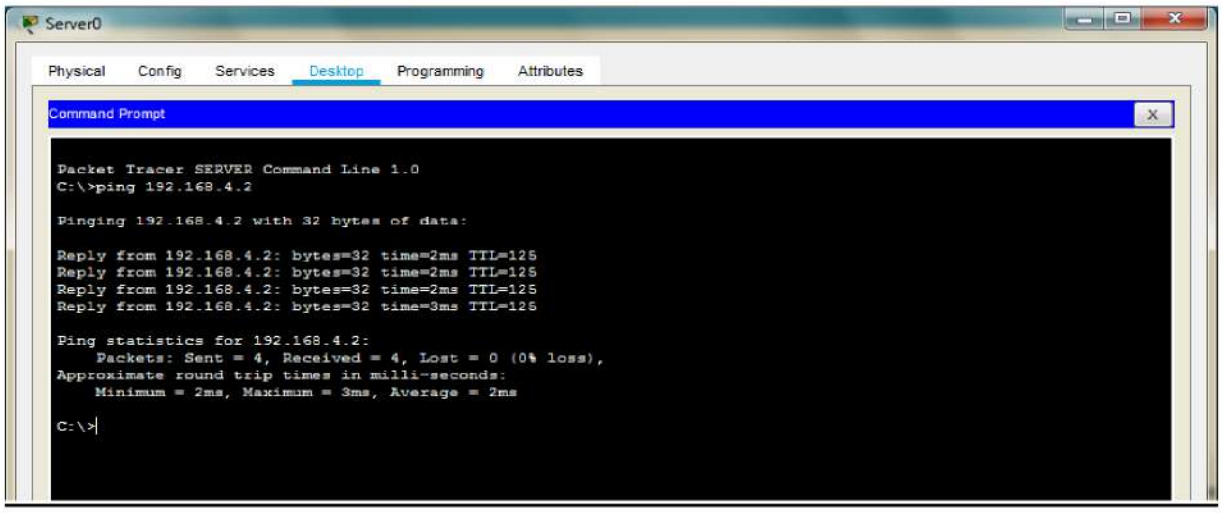


Part 1: Verify Basic Connectivity

We can now verify the connectivity by pinging Server from PC



We can now verify the connecticity by pinging PC from Server



Part 2: Secure Access to Routers

We configure ACL 10 to block all remote access to the Routers and allow remote access only from PC. We type the following commands in all the Routers (Router0, Router1, and Router2). This part is divided in 2 subparts

Part a) Set up the SSH protocol

Enter the following commands in CLI mode of all Routers

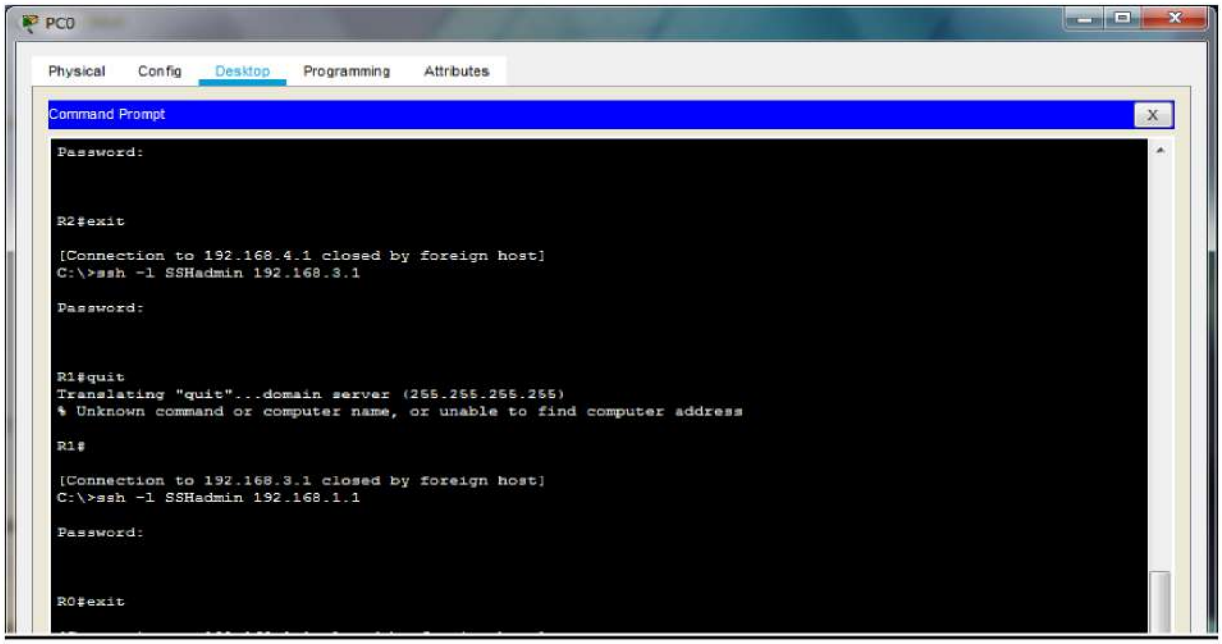
```
Router>enable
Router#configure t
Router(config)#ip domain-name ismail.com
Router(config)#hostname R0
R0(config)#
R0(config)#crypto key generate rsa
R0(config)#line vty 0 4
R0(config-line)#transport input ssh
R0(config-line)#login local
R0(config-line)#exit
R0(config)#username SSHadmin privilege 15 password ismail
R0(config)#exit
R0#
```

Part b) Create an ACL 10 to permit remote access to PC only

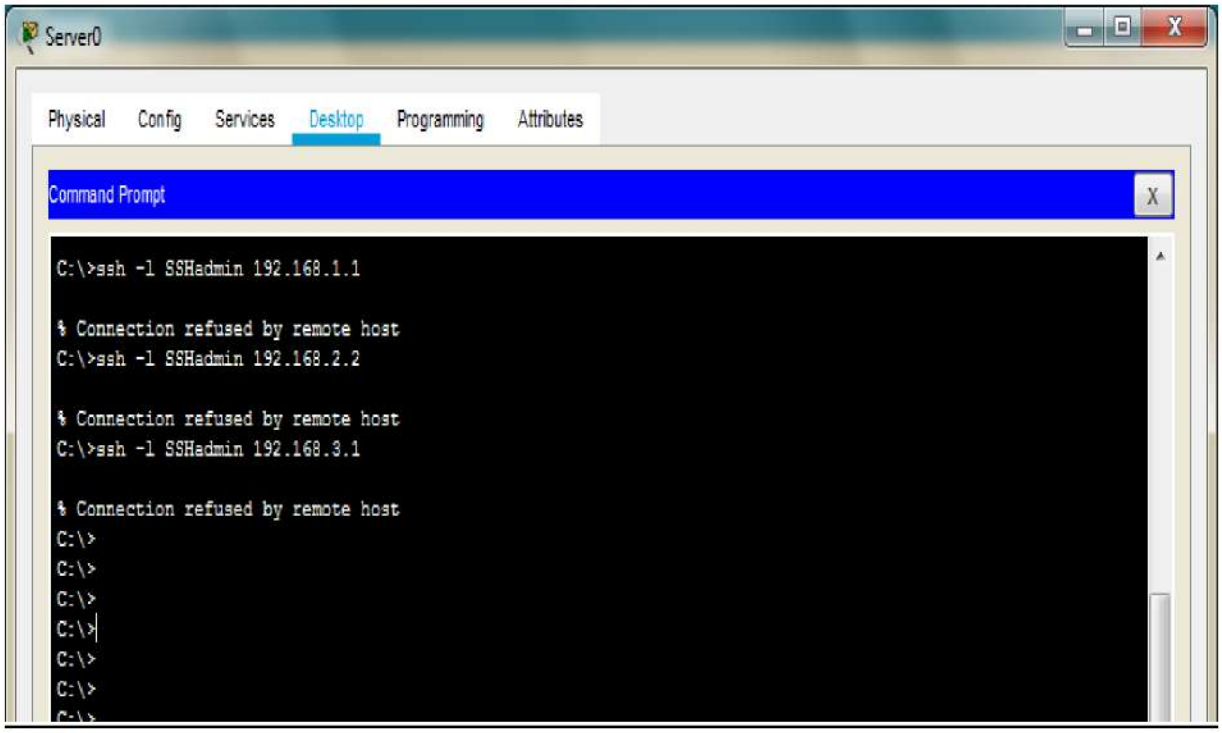
Enter the following commands in CLI mode of all Routers

```
Router>enable
Router#configure terminal
Router(config)#access-list 10 permit host 192.168.4.2
Router(config)#line vty 0 4
Router(config-line)#access-class 10 in
```

Now we verify the remote access from PC using the following and find it to be successful



Now we verify the remote access from Server using the following and find it to be a Failure



Part 3: Create a Numbered IP ACL 120 on R1

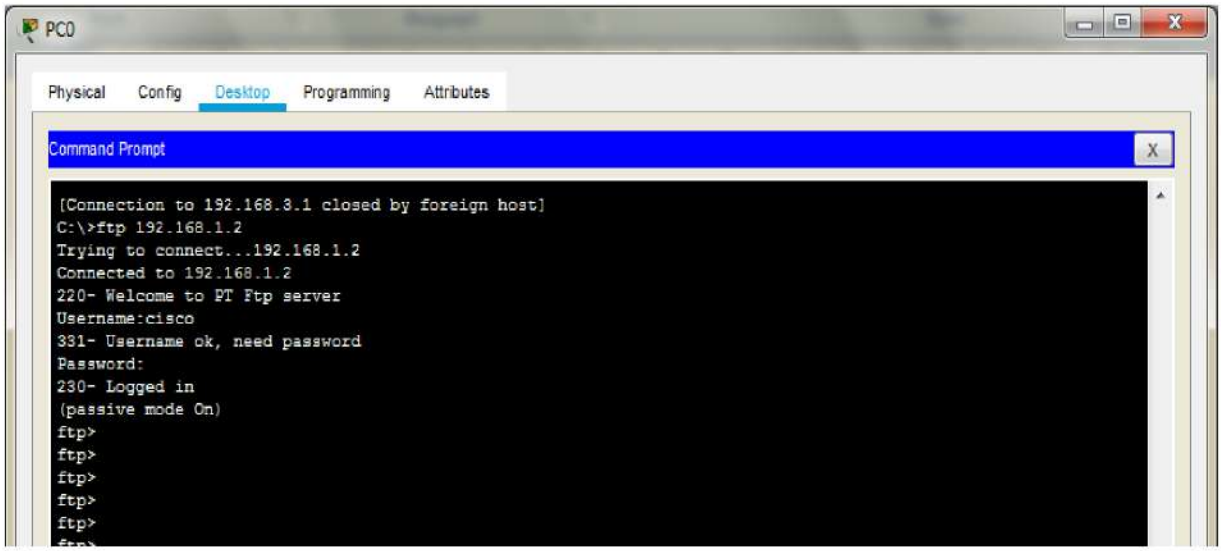
We need to perform the following in this part

1) Create an IP ACL numbered 120 on R1 using the following rules
2) Permit any outside host to access DNS, SMTP, and FTP services on server
3) Deny any outside host access to HTTPS services on **server**
4) Permit **PC to** access **R1** via SSH. (done in previous part)

Enter the following commands in the CLI mode of Router1

```
R1>enable
R1#
R1#configure terminal
R1(config)#access-list 120 permit udp any host 192.168.1.2 eq domain
R1(config)#access-list 120 permit tcp any host 192.168.1.2 eq smtp
R1(config)#access-list 120 permit tcp any host 192.168.1.2 eq ftp
R1(config)#access-list 120 deny tcp any host 192.168.1.2 eq 443
R1(config)#exit
R1#
R1#configure terminal
R1(config)#interface Serial0/1/1
R1(config-if)#ip access-group 120 in
R1(config-if)#exit
```

Verify the above entering the following commands in the PC



Hence we have applied and verified all the required ACLs

For the video demonstration of the above practical scan the QR-code

Configuring IPv6 ACLs

Access Control Lists for IPv6 Traffic Filtering

The standard ACL functionality in IPv6 is similar to standard ACLs in IPv4. Access lists determine what traffic is blocked and what traffic is forwarded at device interfaces and allow filtering based on source and destination addresses, inbound and outbound to a specific interface. Each access list has an implicit deny statement at the end. IPv6 ACLs are defined and their deny and permit conditions are set using the **ipv6 access-list** command with the deny and **permit** keywords in global configuration mode.

IPv6 extended ACLs augments standard IPv6 ACL functionality to support traffic filtering based on IPv6 option headers and optional, upper-layer protocol type information for finer granularity of control (functionality similar to extended ACLs in IPv4).

IPv6 Packet Inspection

Access Class Filtering in IPv6

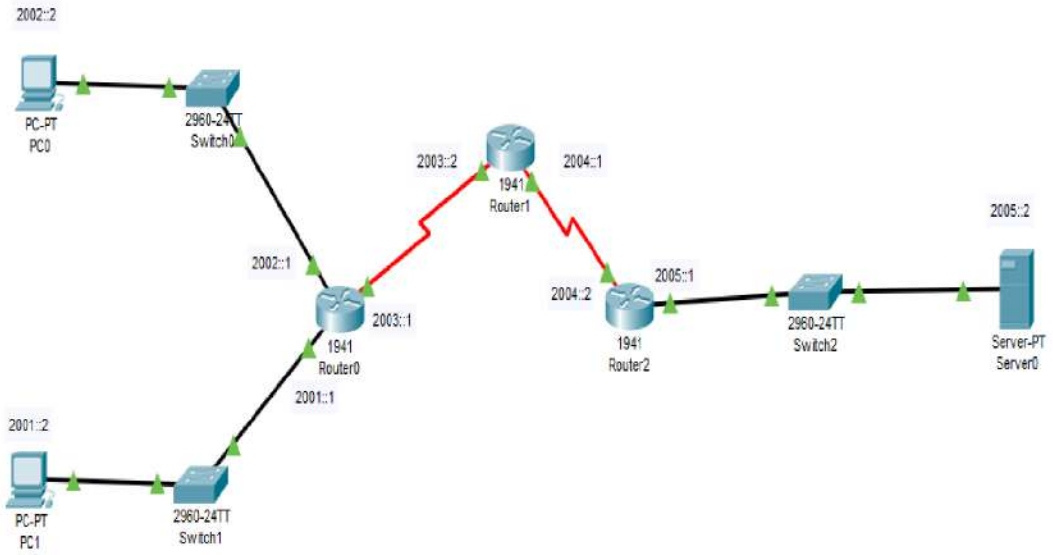
IPv6 Packet Inspection

The following header fields are used for IPv6 inspection: traffic class, flow label, payload length, next header, hop limit, and source or destination IP address. For further information on and descriptions of the IPv6 header fields, see RFC 2474.

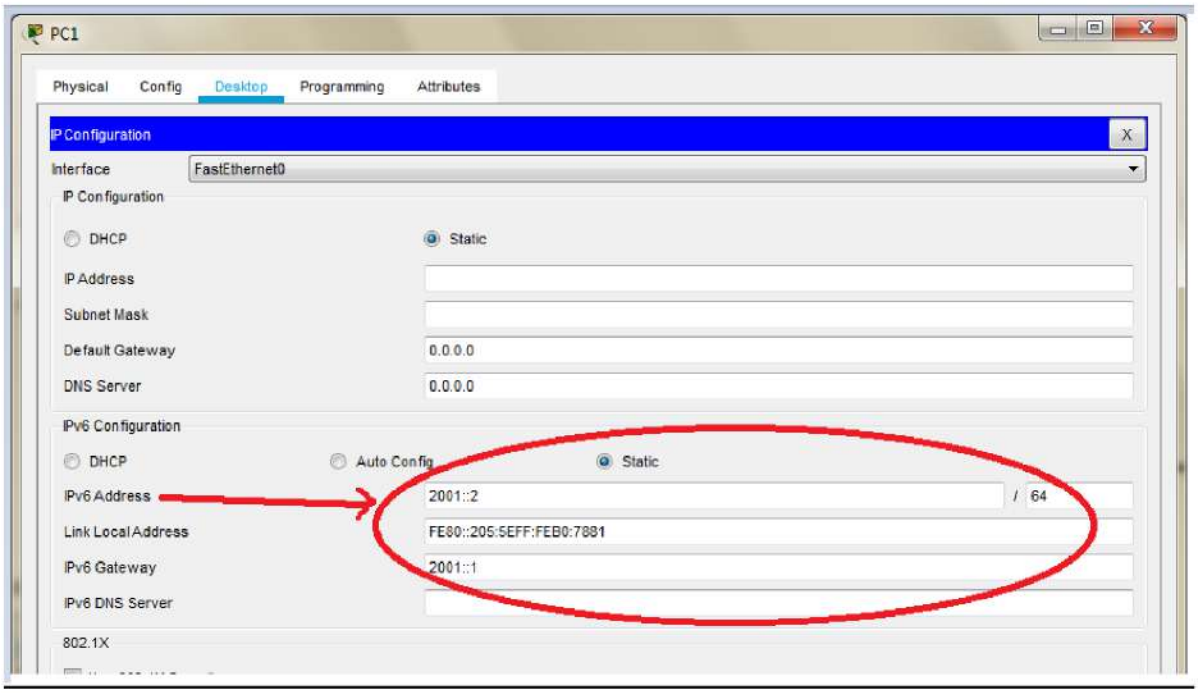
Access Class Filtering in IPv6

Filtering incoming and outgoing connections to and from the device based on an IPv6 ACL is performed using the **ipv6 access-class** command in line configuration mode. The **ipv6 access-class** command is similar to the **access-class** command, except the IPv6 ACLs are defined by a name. If the IPv6 ACL is applied to inbound traffic, the source address in the ACL is matched against the incoming connection source address and the destination address in the ACL is matched against the local device address on the interface. If the IPv6 ACL is applied to outbound traffic, the source address in the ACL is matched against the local device address on the interface and the destination address in the ACL is matched against the outgoing connection source address. We recommend that identical restrictions are set on all the virtual terminal lines because a user can attempt to connect to any of them.

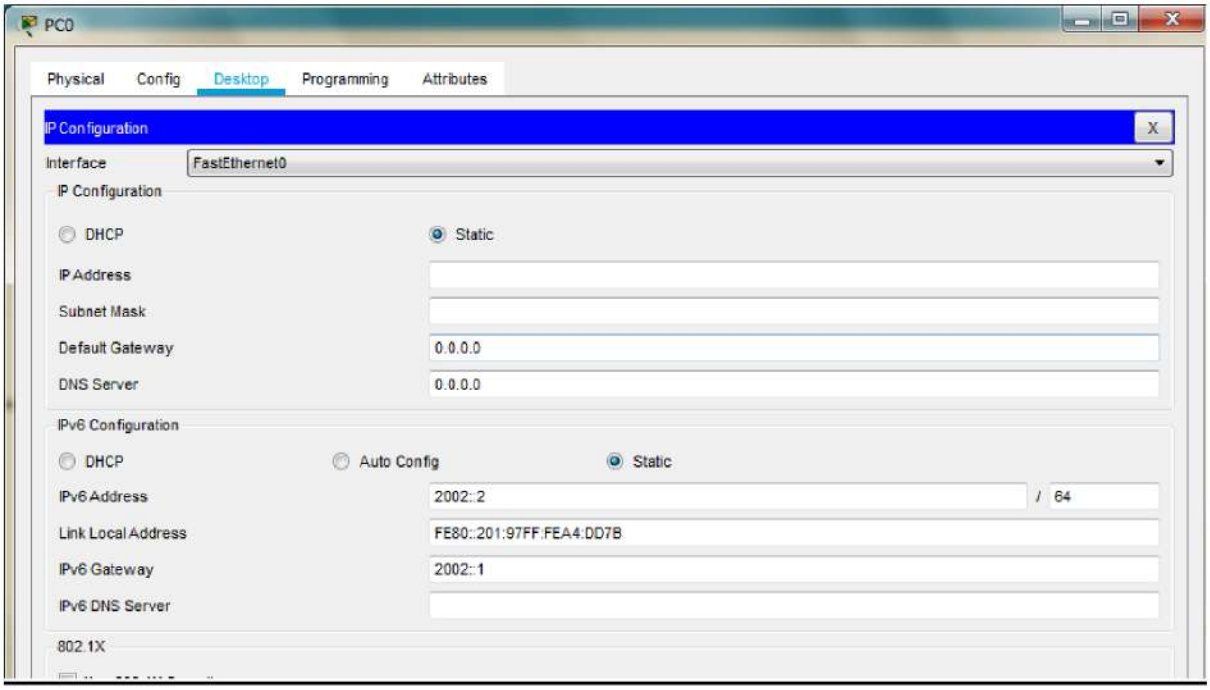
We use the following topology



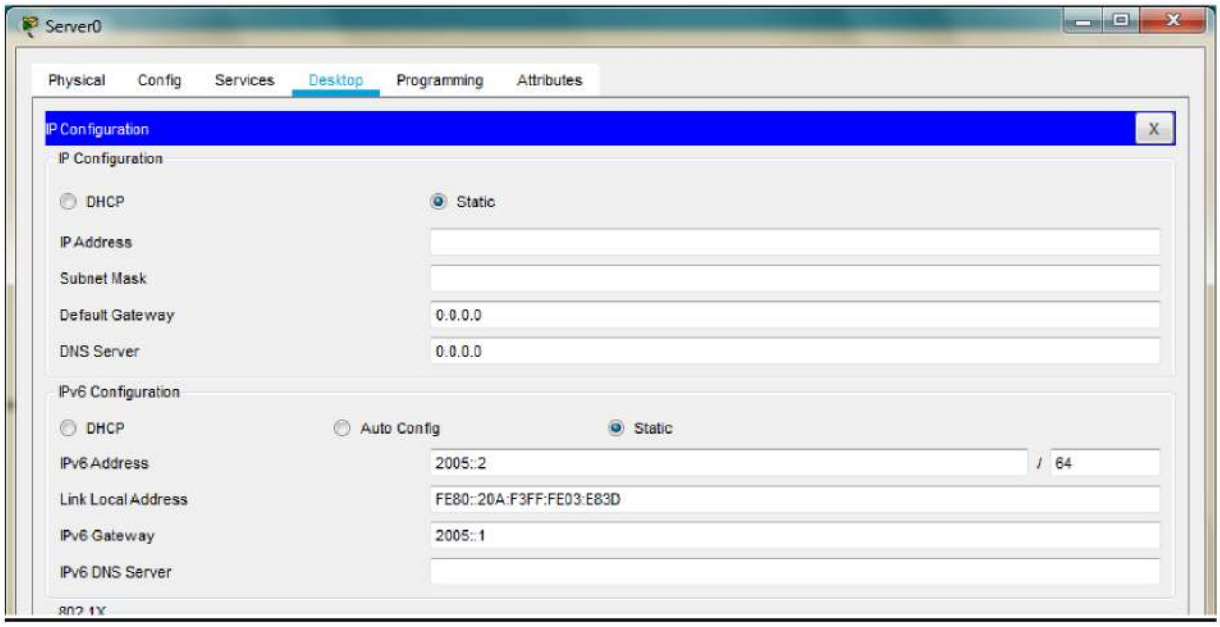
Configuring PC1



Configuring PC0



Configuring Server0



For setting the ipv6 addresses we need to use the CLI mode for each Router as follows

Configuring Router0

```
Router>
Router>enable
Router#
Router#configure terminal
Router(config)#ipv6 unicast-routing

Router(config)#interface GigabitEthernet0/0
Router(config-if)#ipv6 address 2002::1/64
Router(config-if)#ipv6 rip a enable
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#

Router(config)#interface GigabitEthernet0/1
Router(config-if)#ipv6 address 2001::1/64
Router(config-if)#ipv6 rip a enable
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#

Router(config)#interface Serial0/1/0
Router(config-if)#ipv6 address 2003::1/64
```

```
Router(config-if)#ipv6 rip a enable
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
```

Configuring Router1

```
Router>enable
Router#configure terminal
Router(config)#ipv6 unicast-routing
Router(config)#

Router(config)#interface Serial0/1/0
Router(config-if)#ipv6 address 2003::1/64
Router(config-if)#ipv6 rip a enable
Router(config-if)#no shutdown
Router(config-if)#
Router(config-if)#exit
Router(config)#
Router(config)#interface Serial0/1/1
Router(config-if)#ipv6 address 2004::1/64
Router(config-if)#ipv6 rip a enable
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
```

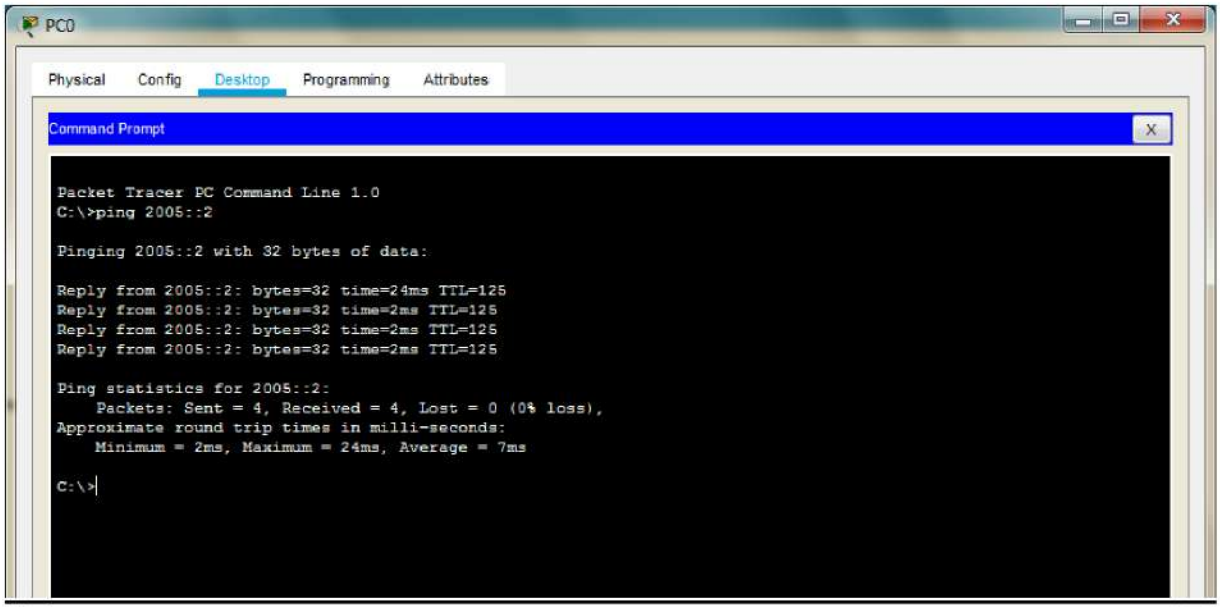
Configuring Router2

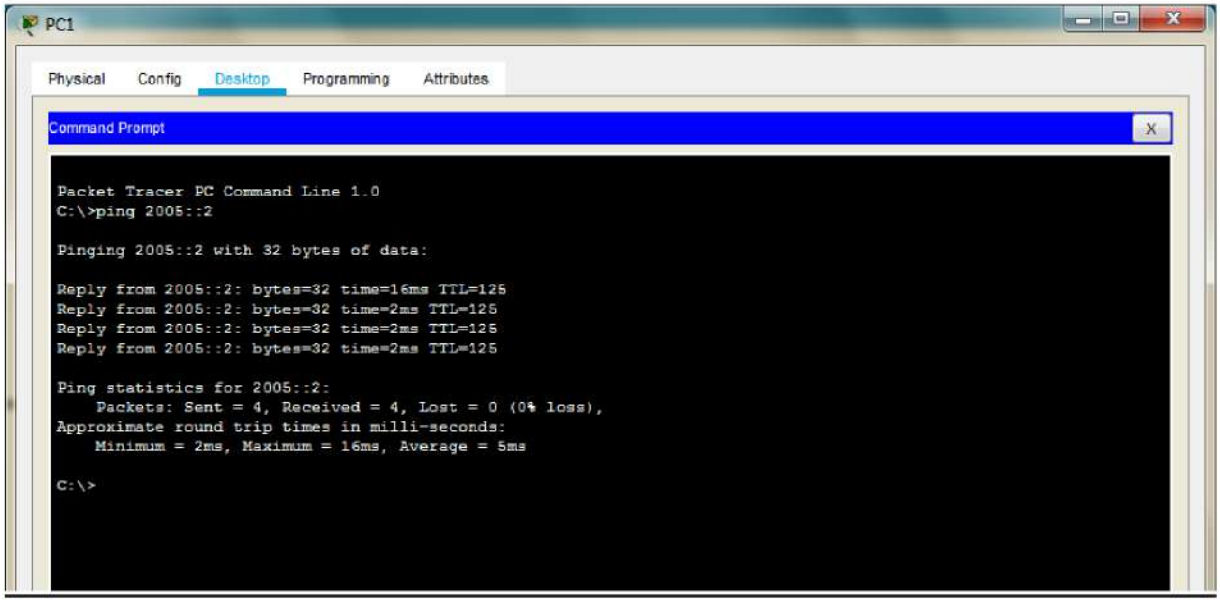
```
Router>enable
Router#configure terminal
Router(config)#ipv6 unicast-routing
Router(config)#

Router(config)#interface Serial0/1/1
Router(config-if)#ipv6 address 2004::2/64
Router(config-if)#ipv6 rip a enable
Router(config-if)#no shutdown
Router(config-if)#exit

Router(config)#interface GigabitEthernet0/0
Router(config-if)#ipv6 address 2005::1/64
Router(config-if)#ipv6 rip a enable
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
```

Check the connectivity by pinging from PCs to Server





And we see that the connectivity is established

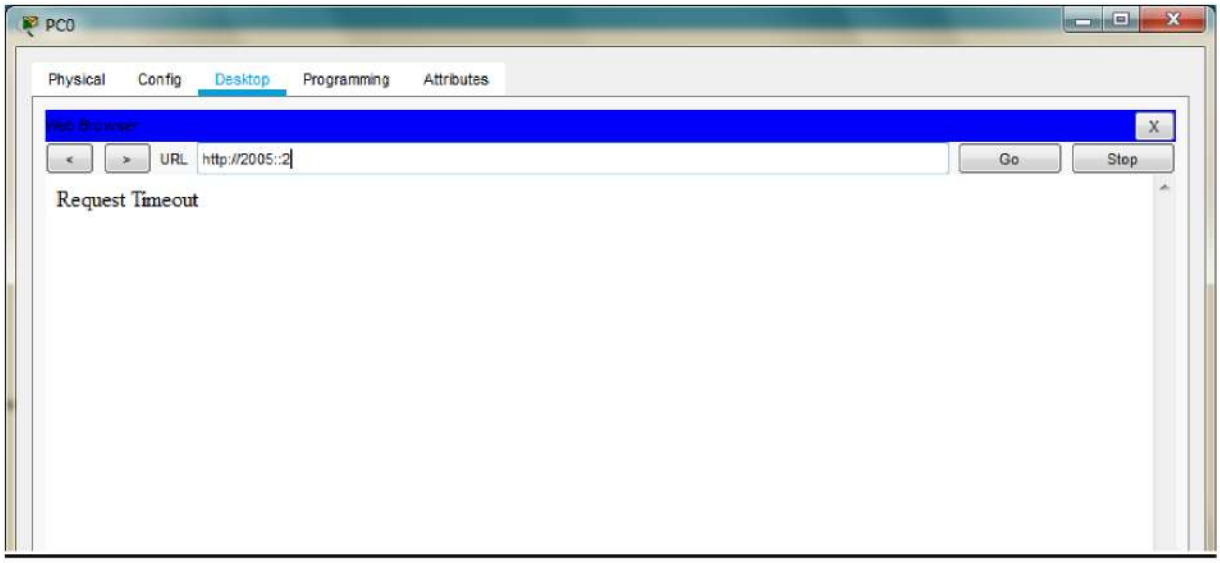
<u>We configure the ACL and apply it to the Router1 with the following conditions</u>

1) No HTTP or HTTPS allowed on server by any host
2) No www service accessible on the server by any host
3) Only ipv6 packets allowed towards the server

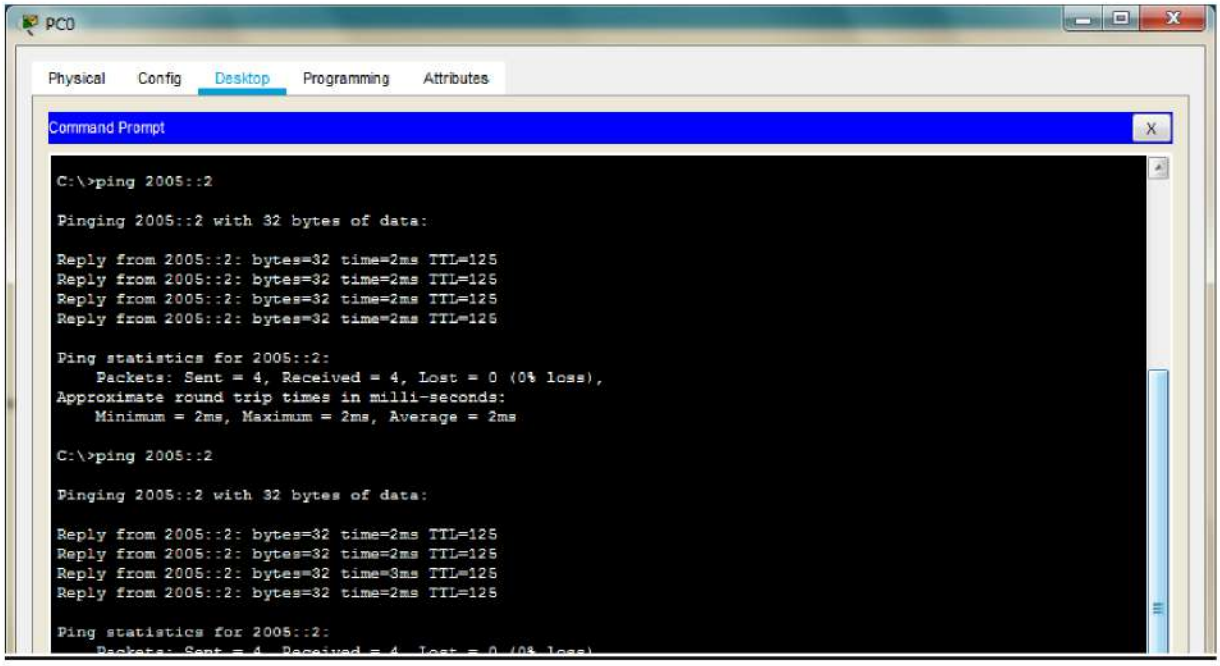
We enter the following commands in the CLI mode of the Router1 and apply it at the proper interface

```
Router>
Router>enable
Router#configure terminal
Router(config)#ipv6 access-list smile
Router(config-ipv6-acl)#deny tcp any host 2005::2 eq www
Router(config-ipv6-acl)#deny tcp any host 2005::2 eq 443
Router(config-ipv6-acl)#permit ipv6 any any
Router(config-ipv6-acl)#
Router(config-ipv6-acl)#exit
Router(config)#
Router(config)#interface Serial0/1/1
Router(config-if)#ipv6 traffic-filter smile in
Router(config-if)#exit
Router(config)#
```

We verify the configuration by first accessing the www service from the browser of both PCs and get failure



Next we verify whether the ipv6 protocol works by pinging server from any of the PC (it must be successful)



Hence the given ACLs have been applied and verified on host running on ipv6 protocol

For the video demonstration of the above practical scan the QR-code